

Nonlinear Dynamics and Chaos - Chapter 2: Flows on the Line

A Geometric Way of Thinking:

Consider $\dot{x} = \sin(x)$ as a flow on the x -axis in the next three problems.

Problem 2.1.1 Find all the fixed points on the flow.

Fixed points occur when $\sin(x) = 0$, so $x = n\pi$ for $n \in \mathbb{Z}$ are the fixed points of this system.

Problem 2.1.2 At which points x does the flow have the greatest velocity to the right?

Since $\sin(x) > 0$ for some points x , the flow to the right is greatest when $\sin(x)$ attains its global maximum. So $x = 2\pi n + \frac{\pi}{2}$ for $n \in \mathbb{Z}$.

Problem 2.1.3a Find the flow's acceleration $\ddot{x}$ as a function of x .

$$\frac{dx}{dt} = \sin(x) \implies \frac{d^2x}{dt^2} = \frac{d}{dt} \sin(x) = \cos(x) \frac{dx}{dt} = \cos(x) \sin(x)$$

Thus we have $\ddot{x} = \sin(x) \cos(x)$.

Problem 2.1.3b Find the points where the flow has maximum positive acceleration.

First, we find the local extrema of $\ddot{x}$ as a function of x .

$$\frac{d}{dx} \ddot{x} = [\cos(x)] \cos(x) + \sin(x)[- \sin(x)] = \cos^2(x) - \sin^2(x)$$

$$\frac{d}{dx} \ddot{x} = 0 \implies \cos^2(x) = \sin^2(x) \implies \cos(x) = \pm \sin(x)$$

The local maximums occur when x satisfies $\cos(x) = \sin(x)$, and the local minimums occur when x satisfies $\cos(x) = -\sin(x)$. We are interested in the former. Thus, $x = n\pi + \frac{\pi}{4}$, for $n \in \mathbb{Z}$. The maximum positive acceleration is $\ddot{x} = \frac{1}{2}$.

Problem 2.4.1a The exact solution of $\dot{x} = \sin(x)$, where $x(0) = x_0$, is:

$$t = \ln \left| \frac{\csc(x_0) + \cot(x_0)}{\csc(x) + \cot(x)} \right|.$$

Given the initial condition $x_0 = \frac{\pi}{4}$, show that

$$x(t) = 2 \tan^{-1} \left(\frac{e^t}{1 + \sqrt{2}} \right).$$

For $x_0 = \frac{\pi}{4}$, we have

$$t = \ln \left| \frac{1 + \sqrt{2}}{\csc(x) + \cot(x)} \right| \implies e^t = \frac{1 + \sqrt{2}}{\csc(x) + \cot(x)} \implies \frac{1}{\csc(x) + \cot(x)} = \frac{e^t}{1 + \sqrt{2}}.$$

So we wish to show that

$$\frac{1}{\csc(x) + \cot(x)} = \tan\left(\frac{x}{2}\right).$$

Recall the double angle identities

$$\sin(2x) = 2 \sin(x) \cos(x),$$

$$\cos(2x) = 2\cos^2(x) - 1.$$

Thus we have

$$\frac{1}{\csc(x) + \cot(x)} = \frac{1}{\frac{1}{\sin(x)} + \frac{\cos(x)}{\sin(x)}} = \frac{\sin(x)}{1 + \cos(x)} = \frac{2\sin(\frac{x}{2})\cos(\frac{x}{2})}{2\cos^2(\frac{x}{2})} = \frac{\sin(\frac{x}{2})}{\cos(\frac{x}{2})} = \tan(\frac{x}{2}).$$

Finally,

$$\tan(\frac{x}{2}) = \frac{e^t}{1 + \sqrt{2}} \implies x(t) = 2 \tan^{-1} \left(\frac{e^t}{1 + \sqrt{2}} \right).$$

Problem 2.1.4b Find the analytical solution for $x(t)$ given an arbitrary initial condition x_0 .

We can simply write

$$x(t) = 2 \tan^{-1} \left(\frac{e^t}{\csc(x_0) + \cot(x_0)} \right).$$

Fixed Points and Stability:

In each of the following equations, find all fixed points, classify their stability, and find an analytic solution, if possible.

Problem 2.2.1 $\dot{x} = 4x^2 - 16$

$\dot{x} = 4(x - 2)(x + 2)$, so the fixed points are $x^* = \pm 2$. Considering the sign of $f'(x^*)$ for each, $x^* = -2$ is stable, and $x^* = 2$ is unstable. We have

$$\begin{aligned} \frac{dx}{dt} = 4x^2 - 16 &\implies \frac{1}{4} \int \frac{dx}{x^2 - 4} = \int dt \implies t = \frac{1}{16} \left(\int \frac{dx}{x - 2} - \int \frac{dx}{x + 2} \right) \\ \implies t = \frac{1}{16} (\ln|x - 2| - \ln|x + 2|) + C &\implies t = \frac{1}{16} \ln \left| \frac{x - 2}{x + 2} \right| + C \end{aligned}$$

Suppose that $x(0) = x_0$. Then we have

$$C = \frac{1}{16} \ln \left| \frac{x_0 + 2}{x_0 - 2} \right|$$

So we have

$$t = \frac{1}{16} \ln \left| \frac{(x_0 + 2)(x - 2)}{(x_0 - 2)(x + 2)} \right|$$

Let $K = \frac{(x_0 - 2)}{(x_0 + 2)}$. Rearranging in terms of x , we get

$$\frac{(x - 2)}{K(x + 2)} = e^{16t} \implies x - 2 = Ke^{16t}(x + 2) \implies (1 - Ke^{16t})x = 2(Ke^{16t} + 1)$$

Thus an explicit solution is

$$x(t) = 2 \left(\frac{Ke^{16t} + 1}{1 - Ke^{16t}} \right)$$

Problem 2.2.7 $\dot{x} = e^x - \cos(x)$

There are an infinite number of fixed points, i.e. solutions to $e^x - \cos(x) = 0$. The rightmost solution is $x^* = 0$.

$$\frac{d}{dx}(e^x - \cos(x)) = e^x + \sin(x)$$

So at $x^* = 0$, $\dot{x} > 0$, and thus $x^* = 0$ is unstable. We can also make some qualitative statements about the infinite number of fixed points for $x^* < 0$. Since $e^x \rightarrow 0$ as $x \rightarrow -\infty$, the fixed points are better approximated by solutions to $\cos(x) = 0$ at $x \rightarrow -\infty$. Furthermore, since $\cos(x)$ oscillates continuously about the x -axis, the stability of these fixed points will alternate between stable and unstable. Since $\frac{d}{dx}e^x \rightarrow 0$ as $x \rightarrow \infty$, and $|\frac{d}{dx}\cos(x)|_{x=x^*} = 1$, the characteristic time scale $\frac{1}{|f'(x^*)|} \rightarrow 1$ as $x \rightarrow \infty$. There is no explicit analytic solution to this differential equation.

Problem 2.2.13 The velocity $v(t)$ of a skydiver falling to the ground is governed by

$$m\dot{v} = mg - kv^2$$

where m is the mass of the skydiver, g is the acceleration due to gravity, and $k > 0$ is a constant related to the amount of air resistance.

a) Obtain the analytic solution for $v(t)$, assuming that $v(0) = 0$.

We have

$$m \frac{dv}{dt} = mg - kv^2 \implies m \int \frac{dv}{mg - kv^2} = \int dt$$

let $u := \sqrt{k}v$. Then $du = \sqrt{k}dv$, and

$$\begin{aligned} \frac{m}{\sqrt{k}} \int \frac{du}{mg - u^2} &= t + C \implies \frac{m}{\sqrt{k}} \cdot \frac{1}{\sqrt{mg}} \left[\tanh^{-1} \left(\frac{u}{\sqrt{mg}} \right) \right] = t + C \\ \implies \sqrt{\frac{m}{kg}} \tanh^{-1} \left(\sqrt{\frac{k}{mg}} v \right) &= t + C \implies \end{aligned}$$

Since $v(0) = 0$, we have $C = 0$. Thus,

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t \right).$$

b) Find the terminal velocity of the skydiver.

$$\lim_{t \rightarrow \infty} \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t \right) = \sqrt{\frac{mg}{k}}.$$

Population Growth:

Problem 2.3.1 There are two ways to solve the logistic equation $\dot{N} = rN(1 - \frac{N}{k})$ analytically for an arbitrary initial condition N_0 [the first is using separation of variables and partial fractions].

b) Make the change of variables $x = \frac{1}{N}$. Then derive and solve the resulting differential equation for x .

Let $x := \frac{1}{N}$. Then we have

$$\frac{dN}{dt} = \frac{dN}{dx} \cdot \frac{dx}{dt} = -\frac{1}{x^2} \cdot \frac{dx}{dt}.$$

Substituting into the logisitic equation, we have

$$-\frac{1}{x^2} \cdot \frac{dx}{dt} = \frac{r}{x} \left(1 - \frac{1}{kx}\right) \implies \frac{dx}{dt} = r \left(\frac{1}{k} - x\right) \implies \int \frac{dx}{x - \frac{1}{k}} = -r \int dt$$

$$\implies \ln \left(x - \frac{1}{k}\right) = -rt + C \implies x = \frac{1}{k} + Ae^{-rt}$$

Since $x(0) = \frac{1}{N_0}$,

$$\frac{1}{N_0} = \frac{1}{k} + A \implies A = \frac{1}{N_0} - \frac{1}{k}$$

Thus,

$$x(t) = \frac{1}{k} + \left(\frac{1}{N_0} - \frac{1}{k}\right)e^{-rt} \implies N(t) = \left(\frac{1}{k} + \left(\frac{1}{N_0} - \frac{1}{k}\right)e^{-rt}\right)^{-1}$$

Linear Stability Analysis:

Problem 2.4.5 Use linear stability analysis to classify the fixed points of $\dot{x} = 1 - e^{-x^2}$.

The only fixed point is $x^* = 0$. We have

$$f'(x) = 2xe^{-x^2} \implies f'(x^*) = 0$$

In order to determine the stability of this point, we need to look at f in the neighbourhood of x^* . For $x = x^* - \varepsilon$, we have $f(x) > 0$. For $x = x^* + \varepsilon$, we have $f(x) > 0$. Thus, this fixed point is bistable (stable from the left, unstable from the right).

Problem 2.4.8 Using linear stability analysis, classify the fixed points of the Gompertz model of tumor growth $\dot{N} = -aN \ln(bN)$, where $a, b > 0$.

The fixed points are $N^* = \{0, \frac{1}{b}\}$. We have

$$f'(N) = [-a] \ln(bN) - aN \left[\frac{1}{bN}[b]\right] = -a(\ln(bN) + 1)$$

so

$$\lim_{N \rightarrow 0^+} f'(N) = +\infty \quad (N^* = 0 \text{ is unstable})$$

$$f'\left(\frac{1}{b}\right) = -a < 0 \quad (N^* = \frac{1}{b} \text{ is stable})$$

Existence and Uniqueness:

Impossibility of Oscillations:

Potentials:

Solving Equations on the Computer: